AT – 3 Analytical Questions

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Deep Learning – CSA4707

Question 1: Training and Validation Loss Curves

A deep learning model is trained over eight epochs. Training loss decreases from 2.0 to 1.0, while validation loss decreases initially from 1.8 to 1.7 and then rises steadily to 4.5. These curves can be used to identify whether the model is learning general patterns or beginning to overfit.

(a) Identify the point at which the model starts to overfit.

Overfitting occurs when a model continues to improve on the training data but performs worse on data it has not seen during training. It often happens when the model learns details or noise specific to the training examples instead of learning patterns that generalize.

From the given values, the validation loss is lowest at Epoch 2, where training loss is 1.8 and validation loss is 1.7. At Epoch 3, training loss decreases to 1.7, but validation loss increases to 1.9. This is the first clear sign that generalization is getting worse while training performance continues to improve.

Epoch 2: Training loss = 1.8, Validation loss = 1.7

Epoch 3: Training loss = 1.7, Validation loss = 1.9

Answer: The model starts to overfit at Epoch 3. Epoch 2 is the best checkpoint based on the lowest validation loss.

(b) Describe a strategy to address overfitting.

A suitable strategy is early stopping. During training, the validation loss is checked after every epoch. The model parameters from the epoch with the lowest validation loss are saved. If validation loss continues to increase for a chosen number of epochs, training is stopped. In this case, the model saved after Epoch 2 would be selected.

Other methods can also help:

1. Dropout: randomly disables a fraction of neurons during training, reducing excessive dependence on particular neurons.

2. L1 or L2 regularization: adds a penalty for large weights, encouraging a simpler model.

3. Data augmentation: creates varied training examples, where appropriate, so the model learns more robust features.

4. More training data: gives the model a broader range of examples and reduces the chance of memorizing a small dataset.

5. Reduce model complexity: use fewer layers or neurons if the network has more capacity than the task requires.

> **Answer:** Use early stopping and restore the model checkpoint from Epoch 2. Dropout, regularization, data augmentation, or reducing model complexity can provide additional protection.

(c) Consequence for unseen data

If training loss keeps decreasing while validation loss keeps increasing, the model is becoming increasingly specialized to the training set. It may memorize training examples and noise rather than learn general features. As a result, its predictions on unseen data may be inaccurate, and its generalization performance will be poor.

> **Answer:** The model is likely to have high error on unseen data because it overfits the training examples.

Question 2: Neural Network Parameters and Parameter Updates

The network is used for binary classification. It has 100 input features, a first hidden layer of 200 ReLU neurons, a second hidden layer of 100 ReLU neurons, and one sigmoid output neuron. The model uses stochastic gradient descent (SGD), a batch size of 32, a learning rate of 0.01, and binary cross-entropy loss.

(a) Calculate the total number of weights and biases.

Every fully connected layer has a weight for each connection between the neurons in two adjacent layers. Each neuron in a layer also has one bias. Therefore:

Number of weights = number of input units × number of neurons in the next layer.

Number of biases = number of neurons in the next layer.

Layer 1: Input layer to first hidden layer

Weights = 100 × 200 = 20,000

Biases = 200

Total parameters = 20,000 + 200 = 20,200

Layer 2: First hidden layer to second hidden layer

Weights = 200 × 100 = 20,000

Biases = 100

Total parameters = 20,000 + 100 = 20,100

Layer 3: Second hidden layer to output layer

Weights = 100 × 1 = 100

Biases = 1

Total parameters = 100 + 1 = 101

Total parameter count

Total = 20,200 + 20,100 + 101 = 40,401

Answer: The network contains 40,401 trainable parameters: 40,100 weights and 301 biases.

(b) Number of parameter updates in one epoch

One epoch means that the model has processed the complete training dataset once. With mini-batch SGD, the model normally performs one parameter update after calculating the gradients for each mini-batch.

Training samples = 10,000

Mini-batch size = 32

Number of batches = 10,000 ÷ 32 = 312.5

A batch cannot contain half a batch, so the final partial batch must also be processed. The number of batches is therefore rounded up.

Number of updates per epoch = ceil(10,000 ÷ 32) = 313

Answer: There are 313 parameter updates in one epoch, assuming the final partial batch is used.

(c) Total updates over 100 epochs

If the same dataset and batch size are used for 100 epochs, the number of updates is the number of updates per epoch multiplied by the number of epochs.

Total updates = 313 × 100 = 31,300

Answer: The model parameters are updated 31,300 times in total.

Question 3: BLEU Score for Machine Translation

BLEU (Bilingual Evaluation Understudy) is an automatic metric for evaluating machine-generated translations. It compares the words and word sequences in a candidate translation with those in one or more reference translations.

Given:

Reference: “je suis étudiant”

Candidate: “je suis étudiant en informatique”

The reference contains 3 words. The candidate contains 5 words. The question requests cumulative BLEU-4 with uniform weights, meaning the precisions for 1-gram, 2-gram, 3-gram, and 4-gram are combined with weights of 1/4 each.

(a)Calculate the cumulative 4-gram BLEU score

Step 1: Calculate unigram precision

Unigrams are individual words. The candidate words are “je”, “suis”, “étudiant”, “en”, and “informatique”. The first three appear in the reference, while “en” and “informatique” do not.

Matching unigrams = 3

Total candidate unigrams = 5

p1 = 3/5 = 0.6

Step 2: Calculate bigram precision

Bigrams are sequences of two consecutive words. The candidate bigrams are “je suis”, “suis étudiant”, “étudiant en”, and “en informatique”. The first two occur in the reference; the last two do not.

Matching bigrams = 2

Total candidate bigrams = 4

p2 = 2/4 = 0.5

Step 3: Calculate trigram precision

Trigrams are sequences of three consecutive words. The candidate trigrams are “je suis étudiant”, “suis étudiant en”, and “étudiant en informatique”. Only “je suis étudiant” occurs in the reference.

Matching trigrams = 1

Total candidate trigrams = 3

p3 = 1/3 ≈ 0.3333

Step 4: Calculate 4-gram precision

A 4-gram is a sequence of four consecutive words. The candidate has two 4-grams: “je suis étudiant en” and “suis étudiant en informatique”. The reference contains only three words, so it has no 4-grams. Therefore, no candidate 4-gram can match the reference.

Matching 4-grams = 0

Total candidate 4-grams = 2

p4 = 0/2 = 0

Step 5: Calculate brevity penalty

BLEU applies a brevity penalty when the candidate translation is shorter than the reference. Here the candidate length c = 5 and reference length r = 3. Since c is greater than r, the brevity penalty is 1.

BP = 1, because $c \geq r$

Step 6: Calculate BLEU-4

With uniform weights, the cumulative BLEU-4 formula is:

$$\text{BLEU-4} = \text{BP} \times \exp[(1/4)\ln(p1) + (1/4)\ln(p2) + (1/4)\ln(p3) + (1/4)\ln(p4)]$$

Because p4 = 0, the product of the four precisions is zero in the standard unsmoothed method. Therefore, the geometric mean is zero.

$$\text{BLEU-4} = 1 \times (0.6 \times 0.5 \times 0.3333 \times 0)^{(1/4)} = 0$$

Answer: The standard unsmoothed sentence-level BLEU-4 score is 0. A smoothing method could produce a non-zero score, but no smoothing method is specified in the question.

(b) Explain BLEU and its suitability

BLEU calculates modified n-gram precision by measuring how many candidate word sequences also appear in the reference translation. It commonly considers n-grams from one word up to four words. It also uses a brevity penalty to discourage translations that are too short.

BLEU is useful because it is fast, reproducible, and can compare translation systems on a large test set without requiring a human to judge every sentence. However, BLEU has limitations: it may penalize valid paraphrases, does not fully measure meaning, and a sentence-level score can be harsh when higher-order n-grams are missing. For this example, the zero score is caused by the absence of matching 4-grams, not necessarily by the translation being entirely meaningless. The standard unsmoothed sentence-level BLEU-4 score is 0 because there are no matching 4-grams. A smoothing method could produce a non-zero score, but no smoothing method is specified in the question.

Question 4: Independent Component Analysis (ICA)

The dataset contains 5,000 chemical compounds, each represented by 1,000 molecular descriptor features. ICA is applied to transform the original features into a smaller set of statistically independent components. The question states that the first 200 components capture 95% of the total variance.

(a) Calculate dimensionality reduction and information retained

The original number of dimensions is 1,000, and the reduced representation contains 200 components.

Original dimensions = 1,000

Reduced dimensions = 200

Dimensions removed = 1,000 − 200 = 800

The percentage reduction is calculated by dividing the number of removed dimensions by the original number of dimensions and multiplying by 100.

Percentage reduction = (800 ÷ 1,000) × 100 = 80%

The question states that the first 200 components capture 95% of the total variance. Using the information provided, the retained information is reported as 95%.

Information/variance retained = 95%

Information/variance not retained = 100% − 95% = 5%

Answer: ICA reduces the feature dimension from 1,000 to 200, an 80% reduction. According to the question, the 200 components retain 95% of the total variance.

Technical note: unlike PCA, ICA does not generally rank components by variance explained. Therefore, the 95% figure should be treated as an assumption supplied by the question rather than a universal property of ICA.

(b) Explain Independent Component Analysis in deep learning

Independent Component Analysis is a statistical technique that attempts to express observed data as combinations of underlying components that are statistically independent. It is often used for feature extraction, signal separation, and reducing redundancy in data.

For a chemical compound dataset, each sample initially has 1,000 molecular descriptors. Some descriptors may contain overlapping or dependent information. ICA transforms these descriptors into a new representation in which the components are designed to be as statistically independent as possible.

A simplified ICA workflow is:

1. Prepare the data: organize the compounds into a matrix and handle missing values or scaling where appropriate.

2. Center and preprocess the features: remove feature means and apply any preprocessing required by the ICA implementation.

3. Estimate independent components: use an ICA algorithm to find a transformation that separates the data into statistically independent components.

4. Select a reduced representation: retain the desired number of components, here 200 as specified in the question.

5. Use the components in a downstream task: provide them to a machine-learning or deep-learning model to predict properties such as biological activity or toxicity.

ICA can be helpful because a lower-dimensional representation may reduce computational cost and can reveal useful latent patterns. However, the extracted components may be harder to interpret directly than the original molecular descriptors, and statistical independence is an assumption that may not perfectly hold in real data.

Answer: ICA transforms correlated observed features into components that are as statistically independent as possible. In this problem, it is used to represent each compound with fewer features before downstream modelling.

Question 5: Deep Boltzmann Machine (DBM)

A Deep Boltzmann Machine is a generative neural network with visible units and multiple layers of hidden units. The given DBM has 100 visible units, 50 units in hidden layer 1, and 50 units in hidden layer 2. It is trained using Contrastive Divergence (CD), with mini-batches of 32 and a learning rate of 0.01.

(a) Calculate the number of weights and biases

Assume the standard DBM architecture in which connections exist only between adjacent layers: visible-to-hidden layer 1 and hidden layer 1-to-hidden layer 2. Each unit has a bias.

Visible layer to hidden layer 1

Weights = $100 \times 50 = 5{,}000$

Hidden layer 1 to hidden layer 2

Weights = $50 \times 50 = 2{,}500$

Bias parameters

Visible biases = 100

Hidden layer 1 biases = 50

Hidden layer 2 biases = 50

Total biases = 100 + 50 + 50 = 200

Total parameters

Total weights = 5,000 + 2,500 = 7,500

Total parameters = 7,500 + 200 = 7,700

Answer: The DBM contains 7,700 parameters under the stated adjacent-layer connectivity assumption.

(b) Calculate memory required using 32-bit storage

Each parameter is stored using 32 bits. There are 7,700 parameters, so the total number of bits is:

Total bits = 7,700 × 32 = 246,400 bits

There are 8 bits in one byte.

Total bytes = 246,400 ÷ 8 = 30,800 bytes

Using decimal megabytes, 1 MB = 1,000,000 bytes.

Memory in MB = 30,800 ÷ 1,000,000 = 0.0308 MB

Using binary units, 1 MiB = 1,048,576 bytes.

Memory in MiB = 30,800 ÷ 1,048,576 ≈ 0.0294 MiB

Answer: The parameters require 30,800 bytes, or 0.0308 MB (approximately 0.0294 MiB). This is only the memory for parameters and does not include gradients, optimizer states, or intermediate training data.

(c) Number of CD iterations for each mini-batch

Contrastive Divergence approximates learning in a Boltzmann machine by comparing statistics from the data with statistics from samples generated by the model. The notation CD-k indicates that k Gibbs sampling steps are performed to generate the model-side sample before estimating the parameter update.

For CD-10, the algorithm performs 10 Gibbs sampling steps for each mini-batch. After these steps, the estimated gradients are used to update the parameters. The 10 Gibbs steps are not 10 separate parameter updates; they are sampling steps within one CD procedure.

Answer: CD-10 performs 10 Gibbs sampling steps for each mini-batch before the parameter update. There is normally one parameter update per mini-batch in this setup.

Conclusion

This assignment covered overfitting and generalization, trainable parameter counting in a neural network, BLEU-based translation evaluation, dimensionality reduction with ICA, and parameter

storage and Contrastive Divergence in a DBM. The numerical results follow the assumptions stated in each question. In particular, the BLEU score uses the standard unsmoothed sentence-level BLEU-4 definition, and the DBM parameter count assumes connections only between adjacent layers.